

## The Flight Of The Nano-Bee

Some time from now—it could be anywhere between five and twenty years—we might or might not see “roboswarms” all around us. Given that, we ask: what will the roboswarms be doing? What will each nanobot be like? And why might we *not* be seeing them? These are pertinent questions, and it is our purpose here to answer them.

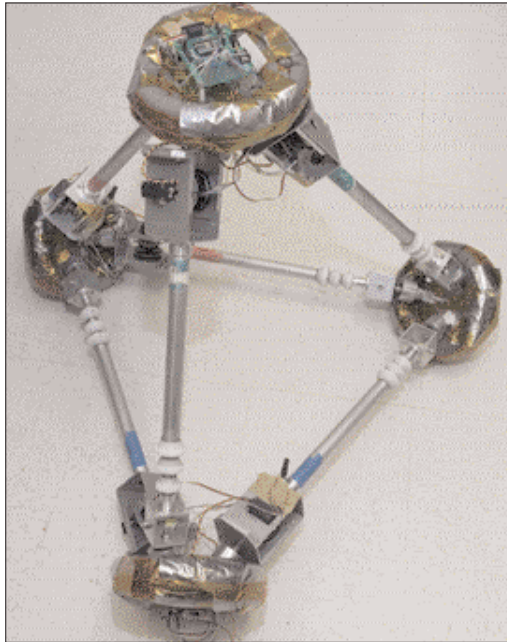
Intelligence in any form is valued by scientists: they notice it, then they study it, then they model it. We’re not talking about artificial intelligence—in fact, we happen to be talking about *natural* intelligence. Mankind has observed a lot of things in nature, including the stars with their patterns, the migration of birds, the evolution of chimpanzees into George W Bush, etc. One such thing was the social behaviour of insects and birds. Ants foraging for food, birds flying in formation, schools of fish swimming and turning together, are all examples of “swarm behaviour”—behaviour that reeks of intelligence, and which has therefore been studied and modelled by computer scientists. The verdict is out: the resulting “swarm intelligence” can be applied to a number of problems and situations, such as optimisation, certain military scenarios, robotics, and more.

That last got us interested, so we decided to investigate.

Robotics and insect swarms taken together would mean, well, useful swarms of insect-like creatures. What might such things do? They could explore the surface of Mars, for a start.

In 1999, the Mars Polar Lander failed. It was, like so many technological edifices today, monolithic: when one thing fails, everything else sinks with it. Instead, imagine a roboswarm trying to explore the planet. It could spread out into different areas: some in the sky, some burrowing into the ground, some cruising along on the surface. If a few of the members of the swarm failed, there would be no problem—does an ant colony shut down shop if a few ants die? The swarmbots would have their sensory apparatus in place, and they would gather data: about soil composition, about temperature, humidity, gases, and so on. Each swarmbot, we’re assuming, will have transmitters in addition to sensors, so all this data would be transmitted to someplace useful—like a central database, which could be on earth. No Mars Lander, no mess, no fuss!

Indeed, NASA has plans. Their Web site talks of a shape-shifting robot called the TETwalker (because it’s a tetrahedron) as follows: “Engineers at NASA’s Goddard Space Flight Center... (success-



The TETwalker robot at NASA’s Goddard Space Flight Center. Such bots will be miniaturised and grouped together, to be deployed as swarms!

fully tested) a shape-shifting robotic pyramid. Robots of this type will eventually be miniaturised and joined together to form ‘Autonomous NanoTechnology Swarms’ (ANTS) that alter their shape to flow over rocky terrain or to create useful structures like solar sails.” We gather that since NASA is talking about shape-altering ANTS, the idea that a roboswarm can do something useful can’t be all stupid.

Nanoswarms are some way into the future, but we’ll see a swarm of *baseballs* tested right this year! July 20, it was reported that a team of MIT researchers led by Prof Steven Dubowsky has conceptualised a swarm of baseball-sized bots to explore the Martian surface. Several thousands of them would explore difficult terrain, going where no probe has gone before—merrily rolling and bouncing away! Each “ballbot” would carry environmental sensors and cameras. They would communicate by forming a LAN, and a base station would relay their data to Earth. A prototype of the swarm will be soon tested on this planet.

## From Darkness Unto Light

We’ve been a little too informal here, so a definition is called for: “Swarm robots are more than just networks of independent agents, they are potentially reconfigurable networks of communicating agents capable of co-ordinated sensing and interaction with the environment.”

How are each of these nanobots—the “agents”—aware of each other, and how do they co-operate for the job they’re doing? That’s precisely where SI comes in. As Yang Liu and Kevin Passino—whom we quoted above—say in *Swarm Intelligence: Literature Overview*:

“The agents use simple local rules to govern their actions and via the interactions of the entire group, the swarm achieves its objectives. ‘Self-organisation’ emerges from the collective actions of the group. SI is the collective intelligence of groups of simple agents... The autonomous agent does not follow commands from a leader, or some global plan. For example, for a bird to participate in a flock, it only adjusts its movements to co-ordinate with the movements of its flock-mates.”

The autonomous agent is the Ant in the Bible, but how *exactly* can intelligent behaviour emerge from simple interactions? It’s complex, but an example should suffice. Think of streams of ants converging towards a speck of food. You’ve seen it: the ants are all moving in various directions. They sometimes go straight, and sometimes they turn. It’s been proved that they’re as likely to turn left as to turn right! Then one ant finds the food. Its duty is to rush back to headquarters and report the find—and herein lies its simple intelligence: it can follow its own trail to go back to where it came from. While doing so, it deposits a “food” pheromone along the way. The next ant that passes the pheromone will follow it—ant instincts dictate that they move in the direction of

“food” pheromone. This second ant returns, and the pheromone along the path is now twice as strong-smelling.

It’s now twice as likely for the next passing ant to take that path, and so on until the entire colony gravitates towards the food the first ant found.

